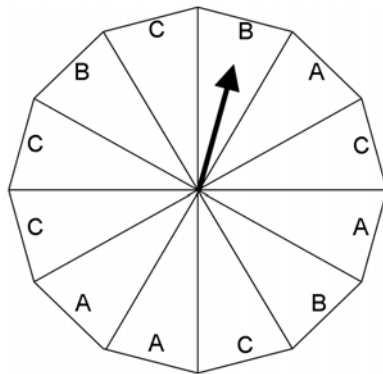


Probability with dice and spinners

Teacher resource

Question 1

A spinner has sections marked as shown. The arrow has an equal chance of stopping on each section.



What is the probability the arrow will stop on B?

$\frac{1}{3}$

☐

3 : 4 : 5

☐

0.25

☐

30%

☐

Answer: C

Skill: Students calculate probability of a simple event.

Background information

Students should understand that probability is a measure of chance or how likely something is to happen. As a measure, the unit chosen is 1 to describe certainty and 0 is used to describe impossibility. The probability of any event must therefore be measured by a number between 0 and 1 inclusive.

For students to manage problems related to probability they must understand the concept of chance itself, the concept of measurement as well as the meaning of fractional numbers expressed as fractions, decimals, ratios and percentages. The difficulty of each of these concepts, coupled with their interaction in a chance situation, makes seemingly simple problems involving probability quite difficult for some secondary students.

Additional questions

- Express this probability as a fraction and percentage.
- Express the probability of the arrow stopping on C as a fraction, percentage and decimal.

Question 2

A fair 10-sided die, with faces marked 1 to 10, is rolled.

Which of the following events has a probability of less than 1?

- ☐ rolling a number greater than 0
- ☐ rolling a number less than or equal to 100
- ☐ rolling a number greater than or equal to 1
- ☐ rolling a number greater than or equal to 10

Answer: D

Skill: Students calculate probability of a simple event.

Additional questions

1. What is the probability of D occurring?
2. Which of the events has a probability of less than one if a fair 6-sided die is rolled?

Question 3

A fair 6-sided die and a fair 10-sided die are rolled. The table below shows how the different totals of the two numbers are calculated.

		Die 1									
		1	2	3	4	5	6	7	8	9	10
Die 2	1	2	3	4	5	6	7	8	9	10	11
	2	3	4	5	6	7	8	9	10	11	12
	3	4	5	6	7	8	9	10	11	12	13
	4	5	6	7	8	9	10	11	12	13	14
	5	6	7	8	9	10	11	12	13	14	15
	6	7	8	9	10	11	12	13	14	15	16

What is the probability of obtaining a total of equal to or greater than 14 with one roll of the two dice?

Answer: 10% or 0.1 or 1/10

Skill: Students calculate probability using data in a familiar context.

Additional questions

1. Express this probability in other ways.
2. What is the probability of rolling a total of equal to or greater than two?

Question 4

A fair six-sided die has faces marked 1, 2, 2, 3, 3 and 3. The die is rolled many times and the number on the top face is recorded. After a number of rolls the number 2 has appeared 20 times.

Indicate the best estimate for the total number of times the die was rolled.

- ☐ 40
- ☐ 60
- ☐ 80
- ☐ 120

Answer: B

Skill: Students apply knowledge of probability to make a prediction.

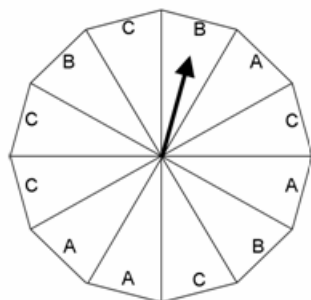
Additional questions

1. What is the probability of obtaining a 3 when this die is rolled?
2. If this die was rolled 600 times, how many times would you expect to get a 2?
3. If after a number of rolls the number 3 has appeared 96 times, how many times would you expect the 2 to have appeared?

Student activity: Probability with dice and spinners

Question 1

A spinner has sections marked as shown. The arrow has an equal chance of stopping on each section.



What is the probability the arrow will stop on B?

$\frac{1}{3}$	3 : 4 : 5	0.25	30%
☐	☐	☐	☐

Question 2

A fair 10-sided die, with faces marked 1 to 10, is rolled.

Which of the following events has a probability of less than 1?

☐	rolling a number greater than 0
☐	rolling a number less than or equal to 100
☐	rolling a number greater than or equal to 1
☐	rolling a number greater than or equal to 10

Question 3

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	4	5	6	7	8	9	10	11	12	13	14
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What is the probability of obtaining a total of equal to or greater than 14 with one roll of the two dice?

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Indicate the best estimate for the total number of times the die was rolled.

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☐	60
☐	80
☐	120